

On the Correctness of the Real Parameterization, the Modular Caveat and the Coefficient Symmetry Compatibility Obstacle

Problem Statement

This text is an explanation for the current version of the article and the corresponding file `GlobalNormalization.v`. Its purpose is to carefully separate four levels of reasoning:

real level,
integer level,
modular level,
coefficient symmetry compatibility level.

At the real level, the parameterization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

is considered, where the parameters

$$m, p$$

are not initially required to be integers. They can be treated as real parameters.

At the integer level, an additional question arises: when is such a real reconstruction compatible with the requirement

$$x, y, z \in \mathbb{N},$$

and, if necessary, with the requirement

$$m, p \in \mathbb{Z}?$$

This is exactly where the parity and exact power conditions appear.

At the modular level, a simple but important methodological caveat is separately fixed: congruence modulo is not the same as equality in integers. Therefore, the existence of solutions to congruences modulo is not in itself a counterexample to the integer formulation of Fermat's Last Theorem.

Finally, the central conditional step of version 15.3 is formulated through the comparison of two natural real normalizations of the same odd binomial core:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n.$$

Here

$$l > 0, \quad q > 0$$

are real residual scales. Comparing these two representations gives the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

In version 15.3, the key point is not the previous one-sided formulation via a separate scale assumption, but the coefficient symmetry compatibility obstacle:

$$l^n = q^n \iff n = 2^{n-1}.$$

In other words, under the two specified normalizations, the equality of the residual scales is possible if and only if the linear binomial coefficient

$$n$$

coincides with the total mass of the odd binomial coefficients

$$2^{n-1}.$$

Next, Coq verifies the elementary final step:

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

Therefore, under the residual scale compatibility requirement, the exponent

$$n > 2$$

is excluded.

It is important to emphasize immediately: this scheme remains conditional. Coq does not prove Fermat's Last Theorem unconditionally. It verifies the links between the formalized premises, the compatibility equivalence

$$l^n = q^n \iff n = 2^{n-1},$$

and the final exponential-linear obstacle. The decisive mathematical point remains the question: must the internal bi-power symmetry of a hypothetical positive integer solution preserve the residual scale equality

$$l^n = q^n$$

between two natural normalizations of the same odd binomial core.

1. Correctness of the Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that it is preliminarily set

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the standard raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

Here it is not required for

$$m, \quad p$$

to be integers. It is sufficient that the expressions are defined in the chosen numerical domain, for example, the domain of real numbers.

However, if it is assumed in advance that

$$x, z \in \mathbb{N},$$

then the equalities

$$x = m^n - p^n, \quad z = m^n + p^n$$

become consistency conditions between the real parameterization and the natural values of

$$x, \quad z.$$

In the real case, there might be no arithmetic restrictions, but in the integer case, additional conditions of parity and exact powers arise.

In Coq, this part is reflected by the lemma

`sum_diff_from_parameters_R,`

which fixes the real identities: if

$$z = m^n + p^n, \quad x = m^n - p^n,$$

then

$$z + x = 2m^n, \quad z - x = 2p^n.$$

At the natural level, the lemma

`parameter_equations_raise_nat`

verifies the elementary fact: if

$$z = m^n + p^n, \quad x = m^n - p^n,$$

are given, then after raising to the power, we automatically obtain

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Thus, at this step, there is no hidden mathematical assumption: only the elementary property of equality is used.

2. Real Factorization of the Odd Binomial Core

Consider the difference

$$(m^n + p^n)^n - (m^n - p^n)^n.$$

According to the binomial formula, only the odd binomial terms remain in it:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S_n(m, p).$$

The first real normalization introduces a positive scale parameter

$$l$$

and writes the core in the form

$$S_n(m, p) = n l^n.$$

Equivalently, if

$$n > 0,$$

then

$$l^n = \frac{S_n(m, p)}{n}.$$

In this sense, the equality

$$y^n = 2n l^n$$

is correct as a real representation.

At the same time, it is not required for

$$l$$

to be an integer. If the positive real case is chosen, then it is assumed that

$$l > 0.$$

In Coq, this is reflected by the definition

$$\text{real_core_factorization } n \text{ core } l,$$

which means

$$0 < n, \quad \text{core} = n l^n$$

over the real numbers. This is not a statement about integer divisibility, but precisely a real factorization.

The lemma

$$\text{real_core_factorization_substitutes_y_power}$$

shows: if

$$y^n = 2 \text{ core}$$

and

$$\text{core} = n l^n,$$

then

$$y^n = 2n l^n.$$

This exactly corresponds to the transition from the odd binomial core to the first scale representation.

3. The Difference Between Real Factorization and Integer Divisibility

If

$$l$$

is considered as a real number, it is not required to assert that the sum

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by

$$n$$

in integers.

Two statements should be strictly distinguished:

$$l^n = \frac{S_n(m, p)}{n}$$

as a real definition, and

$$n \mid S_n(m, p)$$

as an integer arithmetic statement.

The first is permissible in a real model. The second requires a separate proof and is not true in the general case.

For example, for

$$n = 6, \quad m = 1, \quad p = 1$$

the odd sum of binomial coefficients is equal to

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

But

$$32$$

is not divisible by

$$6$$

in integers.

This does not destroy the real scheme because it does not assert the divisibility of the core by

$$n$$

in the ring

$$\mathbb{Z}.$$

The real scheme uses the representation

$$S_n(m, p) = n l^n$$

over

$$\mathbb{R}.$$

That is exactly why the definition

$$\text{real_core_factorization}$$

speaks of representing the core in the form

$$\text{core} = n l^n$$

over real numbers, and not that

$$n$$

is an integer divisor of the core.

4. Integer Specialization and Parity Conditions

If it is additionally required that

$$m, p \in \mathbb{Z},$$

then from

$$z = m^n + p^n, \quad x = m^n - p^n$$

it follows that

$$z + x = 2m^n, \quad z - x = 2p^n.$$

Consequently,

$$z + x$$

and

$$z - x$$

must be even.

In the language of Coq, this is expressed by the lemma

$$\text{sum_diff_from_parameters_Z}$$

and the corollary

$$\text{parity_condition_Z.}$$

If at least one of the parity conditions is violated, then integer parameters

$$m, p \in \mathbb{Z}$$

in such a form are impossible. This is reflected by the lemmas

$$\text{no_parameters_if_parity_violation,} \quad \text{no_parameters_if_odd.}$$

For instance, Coq verifies the specific case

$$z = 2, \quad x = 1, \quad n = 3.$$

Here

$$z + x = 3$$

is odd, so there are no integers

$$m, p$$

such that

$$2 = m^3 + p^3, \quad 1 = m^3 - p^3.$$

This is formalized by the lemma

$$\text{no_parameters_for_example.}$$

Thus, Coq separates the general real level from the integer specialization. The real reconstruction can be formally correct, but the integer reconstruction requires additional arithmetic conditions.

5. Two Real Normalizations of the Odd Binomial Core

The central object of the current scheme is the odd binomial core

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

It arises from the equality

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p).$$

The first normalization isolates the linear binomial factor

$$n$$

and writes

$$S_n(m, p) = n l^n.$$

In Coq, this corresponds to

$$\text{real_core_factorization } n \text{ core } l.$$

This definition means the real factorization

$$\text{core} = n l^n,$$

not a statement about integer divisibility.

The second normalization isolates the total mass of the odd binomial coefficients

$$2^{n-1}.$$

The classical identity has the form

$$\sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} = 2^{n-1}.$$

Therefore, one can introduce a second positive real residual scale

$$q$$

and write

$$S_n(m, p) = 2^{n-1} q^n.$$

Here

$$q$$

is a real scale parameter. It should not be confused with a modulus in modular arithmetic.

In Coq, the second normalization corresponds to the definition

$$\text{coefficient_mass_factorization } n \text{ core } q,$$

which means

$$0 < n, \quad \text{core} = 2^{n-1} q^n$$

over the real numbers.

If the same core is written in two ways,

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

then

$$n l^n = 2^{n-1} q^n.$$

After dividing by

$$n > 0$$

we obtain the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

In Coq, this is verified by the lemma

$$\text{two_scale_factorizations_relation.}$$

Its meaning is as follows: from two real factorizations of the same core,

$$l^n = q^n \frac{2^{n-1}}{n}$$

automatically follows.

This very relation is the bridge to the logic of version 15.3. It shows that the equality of the residual scales

$$l^n = q^n$$

is equivalent to the coincidence of the two extracted coefficient masses:

$$n = 2^{n-1}.$$

6. Coefficient Symmetry Compatibility Obstacle

In version 15.3, the central structural point is not a one-sided hypothesis, but an exact criterion for the compatibility of the two residual scales. The simple equality of residual scales in Coq is defined as

$$\text{residual_scale_equality } n \ l \ q.$$

This means exactly

$$l^n = q^n.$$

The equality of coefficient masses in Coq is defined as

$$\text{coefficient_mass_equality } n.$$

It means

$$n = 2^{n-1}.$$

Comparing the two normalizations gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If the equality of the residual scales

$$l^n = q^n$$

holds, then we get

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since

$$q > 0,$$

we have

$$q^n > 0.$$

Consequently, cancelling

$$q^n$$

is legitimate, and we are left with

$$1 = \frac{2^{n-1}}{n}.$$

Hence

$$n = 2^{n-1}.$$

Conversely, if

$$n = 2^{n-1},$$

then the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}$$

immediately turns into

$$l^n = q^n.$$

Thus, under two real normalizations of the same odd binomial core, we obtain the exact equivalence

$$l^n = q^n \iff n = 2^{n-1}.$$

In Coq, the forward direction is expressed by the lemma

`residual_scale_equality_forces_coefficient_mass_equality.`

It asserts: if the two factorizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

hold, if

$$l^n = q^n$$

holds, and if

$$q > 0,$$

then

$$n = 2^{n-1}.$$

The reverse direction is expressed by the lemma

`coefficient_mass_equality_forces_residual_scale_equality.`

It asserts: if the two factorizations hold and

$$n = 2^{n-1},$$

then

$$l^n = q^n.$$

The full version 15.3 collects both directions into the theorem

`coefficient_symmetry_compatibility_iff,`

that is

$$\text{residual_scale_equality } n \ l \ q \iff \text{coefficient_mass_equality } n.$$

Mathematically, this is exactly

$$l^n = q^n \iff n = 2^{n-1}.$$

For convenience, Coq also introduces the record

`CoefficientSymmetryData.`

This structure packs the data:

$$n, \quad \text{core}, \quad l, \quad q,$$

the positivity

$$l > 0, \quad q > 0,$$

the two factorizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

and the equality of the residual scales

$$l^n = q^n.$$

The lemma

`coefficient_symmetry_data_forces_shift`

derives from this structure

$$n = 2^{n-1}.$$

7. The Elementary Exponential-Linear Obstacle

It remains to analyze the equation

$$n = 2^{n-1}.$$

It is equivalent to

$$2^n = 2n.$$

In positive natural numbers, this equation has exactly two solutions:

$$n = 1, \quad n = 2.$$

Indeed,

$$1 = 2^{1-1} = 1,$$

$$2 = 2^{2-1} = 2.$$

And for any integer

$$n > 2$$

the exponential term

$$2^{n-1}$$

is strictly greater than the linear term

$$n.$$

In Coq, the corresponding part is broken down into several elementary lemmas:

`pow2_gt_linear_shift,`
`pow2_gt_linear,`
`pow_eq_linear_positive,`
`two_pow_eq_two_mul_iff_shift,`
`binary_scaling_roots_only_one_two.`

The final lemma

binary_scaling_roots_only_one_two

asserts:

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

At the level of the packed structure

CoefficientSymmetryData

this yields the lemma

coefficient_symmetry_data_forces_small_exponent,

which asserts:

$$\text{CoefficientSymmetryData} \implies n = 1 \text{ or } n = 2.$$

Finally, the theorem

coefficient_symmetry_data_excludes_high_exponent

says:

$$\text{CoefficientSymmetryData} \implies n > 2 \implies \text{False}.$$

The more general theorem

coefficient_symmetry_compatibility_excludes_high_exponent

formalizes the same final conclusion directly from the two factorizations, the equality of the residual scales, the positivity of

$$q > 0,$$

and the condition

$$n > 2.$$

That is, in the presence of the two factorizations and the residual scale compatibility requirement, an exponent strictly greater than two is excluded.

8. Modular Caveat

This scheme is not a proof via modular arithmetic and does not assert the absence of solutions to the corresponding congruences over finite fields.

The scheme under consideration relates to equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

and not to equations over the finite field

$$\mathbb{F}_{\text{mod } q}.$$

From the integer equality

$$x^n + y^n = z^n$$

a congruence modulo any

$$\text{mod } q$$

indeed follows:

$$x^n + y^n \equiv z^n \pmod{\text{mod } q}.$$

However, the converse transition is generally false. From the congruence

$$x^n + y^n \equiv z^n \pmod{\text{mod } q}$$

the equality

$$x^n + y^n = z^n$$

in integers does not follow.

In other words, a modular congruence only means that there exists an integer

$$t$$

such that

$$z^n = x^n + y^n + \text{mod } q \cdot t.$$

An integer solution to Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions to the congruences

$$x^n + y^n \equiv z^n \pmod{\text{mod } q}$$

for

$$n > 2$$

is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization used in this scheme.

If one specifically transitions to the modular level, then the parameters must be specified separately as residues. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one must separately fix

$$m^n \equiv A \pmod{\text{mod } q}, \quad p^n \equiv B \pmod{\text{mod } q}.$$

After this, the corresponding modular relations take the form

$$z \equiv A + B \pmod{\text{mod } q}, \quad x \equiv A - B \pmod{\text{mod } q}.$$

Thus, the real parameterization and modular arithmetic belong to different levels of description.

9. How the Modular Caveat is Reflected in Coq

In Coq, there is a section

`ModularRemark.`

In it, the relation of modular congruence is defined as

`modular_congruent modq a b.`

Mathematically, this means that there exists an integer

t

such that

$$a = b + \text{modq} \cdot t.$$

The lemma

`integer_equality_implies_modular_power_congruence`

verifies the direction

`integer equality  $\implies$  modular congruence.`

If

$$x^n + y^n = z^n,$$

then one can take

$$t = 0,$$

and obtain the corresponding congruence modulo any

`modq.`

The lemma

`integer_equality_is_zero_modular_witness`

explicitly establishes that a true equality in integers corresponds to the zero modular witness

$$t = 0.$$

The lemma

`modular_congruence_not_integer_equality`

verifies a specific example:

$$3^3 \equiv 1^3 + 1^3 \pmod{5},$$

since

$$27 = 2 + 5 \cdot 5,$$

but at the same time

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers, since

$$2 \neq 27.$$

The structure

`ModularResidues`

formalizes the idea that at the modular level, one must separately specify the residues

$$m^n \equiv A \pmod{\text{mod}q}, \quad p^n \equiv B \pmod{\text{mod}q},$$

and then separately specify

$$z \equiv A + B \pmod{\text{mod}q}, \quad x \equiv A - B \pmod{\text{mod}q}.$$

Thus, Coq strictly confirms the methodological idea:

$$\text{integer equality} \implies \text{modular congruence},$$

but

$$\text{modular congruence} \not\Rightarrow \text{integer equality}.$$

10. General Structure of the Coq Verification

The Coq file verifies the conditional chain of version 15.3. It can be written as follows:

$$\begin{aligned} \text{core} &= n l^n, \\ \text{core} &= 2^{n-1} q^n \end{aligned}$$

imply the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Next, Coq verifies the compatibility equivalence

$$l^n = q^n \iff n = 2^{n-1},$$

and then the elementary arithmetic of natural numbers gives

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

In other words, Coq does not verify a single historical reconstruction in its entirety, but rather the formal correctness of the following bridges:

$$\text{two real factorizations} \implies \text{exact scale relation},$$

$$\text{exact scale relation} \implies (l^n = q^n \iff n = 2^{n-1}),$$

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

The main Coq objects have the following meaning:

- `real_core_factorization` means

$$\text{core} = n l^n$$

over real numbers.

- `coefficient_mass_factorization` means

$$\text{core} = 2^{n-1} q^n$$

over real numbers.

- `two_scale_factorizations_relation` derives from the two factorizations

$$l^n = q^n \frac{2^{n-1}}{n}.$$

- `residual_scale_equality` means

$$l^n = q^n.$$

- `coefficient_mass_equality` means

$$n = 2^{n-1}.$$

- `residual_scale_equality_forces_coefficient_mass_equality` derives from the two factorizations, the equality

$$l^n = q^n$$

and the positivity

$$q > 0$$

the equality

$$n = 2^{n-1}.$$

- `coefficient_mass_equality_forces_residual_scale_equality` derives from the two factorizations and the equality

$$n = 2^{n-1}$$

the equality

$$l^n = q^n.$$

- `coefficient_symmetry_compatibility_iff` verifies the central equivalence of version 15.3:

$$l^n = q^n \iff n = 2^{n-1}.$$

- `CoefficientSymmetryData` packs

$$n, \quad \text{core}, \quad l, \quad q,$$

the positivity

$$l > 0, \quad q > 0,$$

the two factorizations, and the equality of the residual scales.

- `coefficient_symmetry_data_forces_shift` derives from
`CoefficientSymmetryData`

$$n = 2^{n-1}.$$

- `binary_scaling_roots_only_one_two` verifies

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

- `coefficient_symmetry_data_forces_small_exponent` derives from
`CoefficientSymmetryData`

$$n = 1 \text{ or } n = 2.$$

- `coefficient_symmetry_data_excludes_high_exponent` derives from `CoefficientSymmetryData`

$$n > 2 \implies \text{False}.$$

- `coefficient_symmetry_compatibility_excludes_high_exponent` derives the impossibility of the case

$$n > 2$$

directly from the two factorizations, the equality of the residual scales, and the positivity of q .

This structure is fully consistent with the conditional form of the current article version 15.3: the decisive point is not the arithmetic of the equation

$$n = 2^{n-1},$$

but the mathematical status of the residual scale compatibility requirement

$$l^n = q^n$$

between two natural normalizations of the same odd binomial core.

11. What Coq Does Not Prove

For the sake of honesty, it should be explicitly stated that the Coq file does not prove Fermat's Last Theorem unconditionally.

It does not automatically prove that any hypothetical solution to the equation

$$x^n + y^n = z^n$$

for

$$n > 2$$

necessarily implies the preservation of the residual scale equality

$$l^n = q^n$$

between two natural real normalizations of the same odd binomial core.

Coq verifies the following:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n$$

imply

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Furthermore, Coq verifies that under these two normalizations

$$l^n = q^n \iff n = 2^{n-1},$$

and then

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

Consequently, given these premises, an exponent

$$n > 2$$

is impossible.

Thus, the weak point of the scheme is not in the real parameterization, not in the notation

$$\text{core} = n l^n,$$

not in the modular caveat, and not in the Coq-verified final exponential-linear part. The decisive mathematical point remains the question regarding the status of the residual scale compatibility requirement

$$l^n = q^n.$$

12. Consistency with the Current Article

In the current article version 15.3, the main conditional result is formulated as follows: if a hypothetical positive natural solution to Fermat's equation

$$x^n + y^n = z^n, \quad n > 2$$

admits the considered symmetric real parameterization, and if the same odd binomial core admits two natural real normalizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

then their comparison gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Moreover, the equality of the residual scales

$$l^n = q^n$$

is equivalent to the equality

$$n = 2^{n-1}.$$

But this equality is only possible for

$$n = 1 \quad \text{and} \quad n = 2.$$

Therefore, the exponent

$$n > 2$$

is excluded under the specified residual scale compatibility requirement.

In this sense, the Coq file matches the article: it verifies exactly the final formal chain

$$\begin{aligned} \text{core} &= n l^n, \\ \text{core} &= 2^{n-1} q^n, \\ l^n &= q^n \frac{2^{n-1}}{n}, \\ l^n = q^n &\iff n = 2^{n-1}, \\ n = 2^{n-1} &\implies n \in \{1, 2\}. \end{aligned}$$

At the same time, the modular caveat remains a separate methodological block: it does not participate in the proof of the final exponential-linear obstacle, but protects the reasoning from mixing different levels — integer equalities and modular congruences.

Conditional Theorem. Let, for some natural exponent

$$n > 0$$

the same odd binomial core admit two real factorizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

where

$$l > 0, \quad q > 0.$$

Then their comparison gives

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Consequently,

$$l^n = q^n \iff n = 2^{n-1}.$$

Since

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2,$$

then under the residual scale compatibility requirement the case

$$n > 2$$

is impossible.

Final Conclusion. Understanding that

$$m, p, l, q \in \mathbb{R}$$

the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

as well as the real factorizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n$$

are correct as a formal real scheme.

The Coq file consistently formalizes this scheme, separately fixes the real factorization, integer specialization, modular caveat, two normalizations of the odd binomial core, the compatibility relation, and the final exponential-linear comparison.

The main verified chain has the form

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n$$

$$\implies$$

$$l^n = q^n \frac{2^{n-1}}{n}$$

$$\implies$$

$$l^n = q^n \iff n = 2^{n-1}$$

$$\implies$$

$$n \in \{1, 2\}.$$

Consequently, under the residual scale compatibility requirement, exponents

$$n > 2$$

are excluded.

Modular congruences for

$$n > 2$$

may exist, but they belong to a different level of the problem and are not integer solutions to Fermat's equation.

Thus, the final status of the scheme of version 15.3 is clear: the final Coq-verified part is correct as a conditional chain, and the decisive mathematical question is focused on justifying whether the internal bi-power symmetry of a hypothetical positive integer solution must preserve the residual scale equality

$$l^n = q^n$$

between two natural normalizations of the same odd binomial core.